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PREMIUM ADDENDUM — Advanced Analytics (measured)

A. Convergence analytics

Nearest singularity of the DELIVERED trajectory (exact 40-digit first-integral quadrature on the delivered slope): $x = 0.998387$.

B. Conditioning indicators (measured)

step_width_min = 0.015625

step_width_max = 0.015625

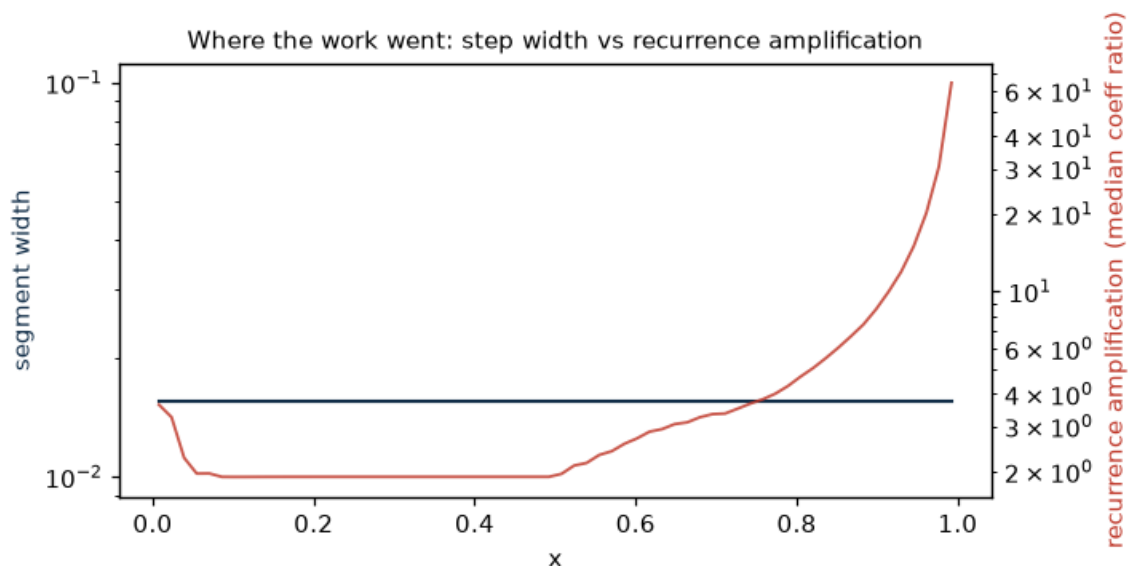
stiffness_indicator = 1

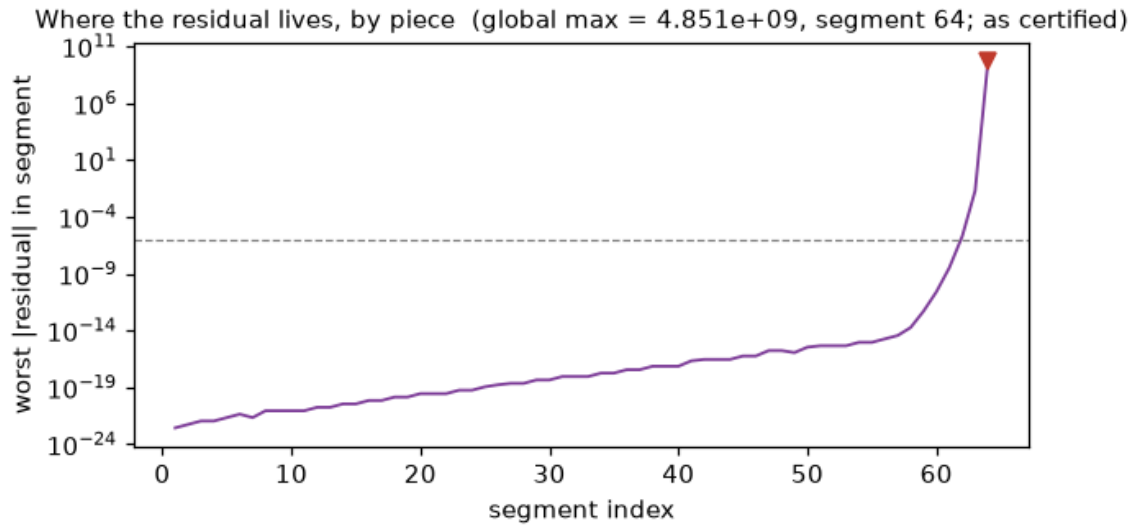
recurrence_amplification_median = 2.21207

recurrence_amplification_max = 64.5568

Method: measured from this run's own pieces: adaptive step widths (stiffness = max/min width) and coefficient-ratio growth of the recurrence; indicators, not rigorous condition numbers.

Figure — step width vs recurrence amplification





C. Perturbation sensitivity

Available as a custom add-on module - contact LQI to enable. Perturbation sensitivity re-solves the problem with perturbed data to measure how far the answer moves; it is priced as a Premium module and was not part of this order.

D. Recurrence (LaTeX export)

```
\begin{aligned}
u_{1} &:= \sinh{\left(20\ p_{0}\ \right)} \ \backslash\backslash \\
u_{2} &:= \cosh{\left(20\ p_{0}\ \right)} \ \backslash\backslash \\
p_{0}' &= p_{1} \ \backslash\backslash \\
p_{1}' &= 20\ u_{1} \ \backslash\backslash \\
u_{1}' &= 20\ p_{1}\ u_{2} \ \backslash\backslash \\
u_{2}' &= 20\ p_{1}\ u_{1} \\
\end{aligned}
```

% coefficients follow the Cauchy-product recurrence of this polynomial system about each segment's left node.

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